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SURVEY

A Recent Survey of Vision Transformers for Medical Image Segmentation

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ABSTRACT Medical image segmentation plays a crucial role in various healthcare applications, enabling accurate diagnosis, treatment planning, and disease monitoring. Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in this domain due to their proficiency in learning complex patterns from raw data. In recent years, Vision Transformers (ViTs) have gained significant attention as an effective approach for various challenges in image analysis. However, they may lack image-related inductive bias and translational invariance that may affect their performance. To address this, Hybrid Vision Transformers (HVTs) have been introduced, combining CNNs and Transformer layers to effectively analyze features at both local and global scales. Building on the success of ViTs and HVTs, this paper reviews recent advancements in these architectures for medical image segmentation. We classify approaches based on architectural design and review state-of-the-art models for different imaging modalities, analyzing their limitations and potential solutions. Additionally, we highlight key challenges, discuss current trends and propose future research directions in the field. This review aims to provide valuable insights for researchers and professionals working on ViT-based medical image segmentation.

INDEX TERMS Vision transformers (ViTs), multi-head self-attention, hybrid ViTs (HVTs), medical image segmentation, deep learning.

I. INTRODUCTION

In medical image analysis, image segmentation has a significant role [1]. Accurate and efficient segmentation in medical images can help physicians in the identification and evaluation of specific anatomical characteristics, diseases, or regions of interest. These segmented regions are then analyzed to estimate the disease prognosis and to devise effective treatment strategies [2].

Over the past ten years, deep learning techniques have driven significant advancements in medical image segmentation, more especially Convolutional Neural Networks (CNNs) because of their capacity to capture intricate patterns from unprocessed data [3]. CNNs tend to capture the local

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TABLE 1. Table of abbreviations.

Abbreviation	Full form
AP	Average Precision
ATEM	Adaptive Token Extraction Module
AUC	Area Under the Curve
CCA	Channel Cross-Attention
CNN	Convolutional Neural Network
CT	Computed Tomography
DCA	Dual Cross-Attention
DSC	Dice Score
EDformer	Encoder-Decoder Transformer
ET	Enhancing Tumor
FCN	Fully Convolutional Network
FLOPs	Floating Point Operations
GANs	Generative Adversarial Networks
GT U-Net	Group Transformer Network
HVT	Hybrid Vision Transformer
IoU	Intersection over Union
Max-SA	Multi-Axis Self-Attention
MaxViT	Multi-Axis Attention-based Vision Transformer
MAP	Mean Average Precision
MBConv	Mobile inverted bottleneck convolution
MIS	Medical Image Segmentation
MRI	Magnetic Resonance Imaging
MS-DMSA	Multi-Scale Deformable Multi-Head Self-Attention
MSA	Multi-head self-attention
NLP	Natural language processing
PVT	Pyramid Vision Transformer
RAT-Net	Region-Aware Transformer Network
RPA	Reciprocal Attention Module
SCA	Spatial Cross-Attention
SFRM	Selective Feature Reinforcement Module
Swin	Shifted Window
TC	Tumor Core
UNETR	UNet Transformer
ViT	Vision transformer
WSI	Whole Slide Image
WT	Whole Tumor

correlations in the images, which allows them to learn the local patterns distributed among the entire image [4]. Their inherent nature to extract 2D features from images gave them an edge over traditional methods. The UNet is one of the first CNN-based encoder-decoder techniques that revolutionized medical image segmentation tasks [5]. Over time many UNet-like algorithms were proposed for various medical image segmentation tasks, that demonstrated remarkable improvement in various medical image modalities [6], [7], [8]. However, their capacity to model global relationships in the images is constrained by the local nature of the convolution operator [9].

Researchers have proposed numerous approaches to incorporate global-level information into the CNN-based segmentation architectures [10], [11]. In this regard, various attention-based approaches were developed to focus on the important regions in the medical images [12]. Recently, Transformers which were originally introduced for Natural Language Processing (NLP) [13], have gained prominence in various image-related problems, including computer vision and medical image analysis [14], [15].

The multi-head self-attention (MSA) mechanism in Vision Transformers (ViTs) possesses the capability to model global relationships within the images [14]. These ViT-based architectures have demonstrated remarkable results in handling complex medical image segmentation tasks in various image modalities. The inherent ability of ViTs to model global features from medical images involves initially dividing them into patches and subsequently processing each patch through their self-attention mechanism [16]. However, ViT architecture does not include image-related inductive bias to deal with the positional variations in the images. In addition, their performance may be affected by the amount of training data, as ViT models generally require large training data for learning [17].

Recently, there has been a growing interest in integrating CNN inductive biases into ViT architectures, leading to the emergence of CNN-Transformers, also known as Hybrid Vision Transformers (HVTs) [18]. These CNN-Transformer architectures combine the MSA mechanism of ViTs with the convolutional operations of CNNs to model both local and global relationships within images. Several novel frameworks have been developed, leveraging CNN-Transformer architectures and showcasing impressive results in tasks related to medical image segmentation [19], [20], [21].

With the rapid advancement of medical image segmentation techniques utilizing ViTs, various interesting surveys have been conducted to review these approaches [7], [11], [15], [18]. Most of these surveys are either focused on a special organ or a special modality. This paper presents a comprehensive survey that spans recent ViT-based segmentation approaches applied to medical images across diverse modalities. In addition, we broadly categorize these methods into 1) pure ViT-based methods, and 2) HVT-based methods based on their architectural design. Secondly, we devise a taxonomy for HVT as a) encoder-based integration, b) decoder-based integration, and c) integration at the encoder-decoder interface.

The goal of this study is to give a comprehensive overview of the latest techniques, strategies, and developments related to the utilization of ViTs for medical image segmentation. We intend to provide researchers, clinicians, etc. with an extensive comparison between the existing ViT architectures developed for precise and effective medical image segmentation.

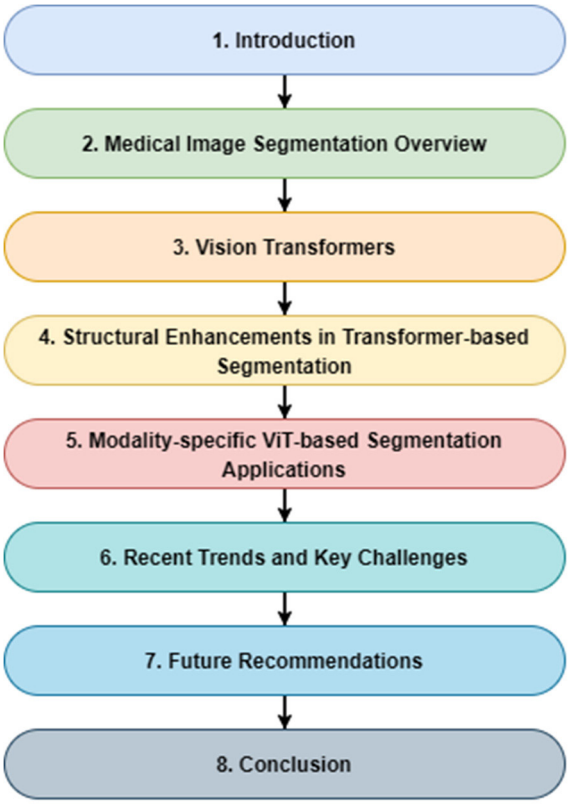


FIGURE 1. Overall structure of the survey and section map.

The rest of this survey is organized as follows (shown in Figure 1): The basic ideas behind medical image segmentation and ViTs are summarized in Section II and Section III, respectively. Section IV provides a thorough analysis of ViT and HVT-based medical image segmentation techniques. We categorize these approaches based on the utilization of ViT and HVT in their encoder-decoder-based architecture. Section V provides a detailed overview of ViT and HVT-based methods for medical image processing across various modalities, including CT, histopathology, MRI, ultrasound, and X-Ray. Section VI summarizes several techniques utilized by HVT-based medical image segmentation approaches to enhance their performances. Section VII presents challenges and future recommendations, and Section VIII concludes the survey.

II. MEDICAL IMAGE SEGMENTATION OVERVIEW

Diagnosis and treatment of diseases and other medical conditions require medical tests that help physicians see inside a body. In this regard, medical imaging is essential since it allows for non-invasive viewing and diagnosis of a variety of medical disorders [22]. Different imaging modalities offer unique insights into different aspects of the human body [23]. Medical images are of great importance in diagnosing and treating diseases as they provide valuable information about the symptoms. However, when examining medical images at the pixel level, only specific regions within these images are relevant, such as the tumor region in a CT scan.

Accurate medical image segmentation is vital for computer-assisted diagnosis, image-guided procedures, and therapeutic planning [24], [25], [26]. Medical image segmentation involves segmenting specific areas of interest, such as infected regions or abnormal organs, from medical images [23], [27]. However, this process poses several challenges for researchers [28], including color and quality variations due to differences in image acquisition methods, heterogeneity in the anatomical structures, class imbalance problem, unavailability of labelled data, presence of noise and artifacts, and high pattern diversity within a single image. These factors significantly complicate the accurate segmentation of the object of interest.

In clinical pathology, the gold standard is to carry out manual medical image analysis that involves visual examination of prepared slides under a light microscope, and manual demarcation of regions of interest, such as tumors, lesions, or specific anatomical structures by highly trained pathologists [29]. Moreover, this manual assessment is tedious and exhausting, often influenced by human subjectivity, and requires expert clinicians [30]. However, digital pathology has introduced a new paradigm shift, enabling a fast and accurate diagnosis by providing researchers with the ability to design automated systems. These automated systems can help pathologists by providing them with a fast diagnosis that can be used as a second opinion, thus reducing their burden [31]. Different modalities in medical imaging depict the human body's internal structures. Segmentation applied to these modalities can isolate specific organs as well as aid in disease identification within a specific organ.

For automated diagnosis, several traditional methods have been applied for the segmentation of medical images over the years. These methods encompass a range of techniques, including region-based segmentation techniques [32], edge detection techniques [33], statistical shape models [34], [35], active contour models [36], Thresholding [37], clustering [38], and machine learning methods [39]. However, efficient automated medical image segmentation is vital for clinical assessment but remains challenging due to (1) significant variations in object sizes, (2) ambiguous boundaries and diverse textural variations, and (3) low-intensity contrast when isolating objects from the background [25], [26].

To overcome the challenges posed by traditional medical image segmentation methods, researchers widely utilized deep learning-driven methods, particularly CNNs. These methods demonstrated considerable performance in various medical image segmentation approaches [40], [41]. Deep learning-based segmentation algorithms tend to perform well when they are provided with dense pixel-level annotations. CNNs are effectively utilized for a variety of applications, such as tumor detection [42], [43], COVID-19 detection [44], skin lesion detection [45], and segmentation [46], [47] in various imaging modalities including, ultrasound, MRI, X-ray, CT, microscopy and others. U-Net, due to its use of skip connections, became highly popular and demonstrated exceptional performance across various medical image segmentation tasks [48]. A variety of medical image segmentation approaches utilized U-Net-based architecture to effectively segment the region of interest [48]. However, CNNs may struggle to learn explicit long-distance dependencies because of their limited receptive fields [9]. Therefore, recently researchers have come up with ViT-based and hybrid (utilizing both CNN and ViT) methods to solve complex medical segmentation problems [49], [50]. These architectures are discussed in the later sections.

III. VISION TRANSFORMERS

Over the past decade, CNNs have been extensively utilized for a variety of computer vision tasks, such as the analysis of medical images. However, the limited receptive field of convolution filters restricts CNNs from capturing prolonged dependencies within medical images [51], [52] which may affect their performance.

Recently, Transformers have gained significant attention due to their outstanding performance. Initially proposed for machine translation, Transformers became popular in NLP tasks because of their effective sequence processing through the self-attention mechanism. Inspired by the success of Transformers in NLP, [14] proposed ViT, which processes images as a sequence of tokens rather than as a 2D grid of pixels. By converting the input image pixels into sequences and using a self-attention mechanism to identify long-range dependencies and interactions between the image parts, ViT can recognize the intrinsic structures that exist in images [53].

ViT divides the input image into distinct, non-overlapping patches of fixed size, converting each patch into a feature representation called a patch embedding. These embeddings are linearly transformed and serve as input tokens for the ViT encoder [54], allowing the model to capture the global context of the entire image. To incorporate positional information, spatial embeddings are added to the patch representations [55], enabling the model to maintain awareness of the spatial configuration of patches in the original image.

The encoder, a fundamental component of the ViT architecture, comprises multiple layers of MSA and feed-forward neural networks. The MSA heads capture both local and global relationships among patches, while the feed-forward

networks introduce non-linearity to refine features [56]. The output of the final encoder layer is typically pooled, and a classification head is added for predictions. For image classification tasks, this pooled representation is followed by a fully connected layer with the softmax activation to produce class probabilities. In contrast, for image segmentation, a specialized decoder block generates image masks [57].

ViTs are effective for medical images because they can learn global context. This capability leads to significant performance improvements, especially in organs with large receptive fields, such as the lungs. The self-attention mechanism in ViTs allows for the modeling of intricate relationships between image patches, irrespective of their spatial proximity. In contrast, CNNs focus primarily on local correlations within images because of their limited receptive fields. Numerous ViT-based systems have been developed for diverse medical image-related applications [58], [59], [60], [61], [62], [63].

ViT-based medical image segmentation methods have shown exemplary performance in various organs, including brain tumor segmentation [64], [65], [66], cell nuclei segmentation [67], [68], colon cancer segmentation [69], liver lesion segmentation [70], kidney tumor segmentation [71], multiple-organ segmentation [72], and skin lesion segmentation [73], [74]. The impressive performance of ViT architectures in these tasks has motivated researchers to enhance them further by improving position encoding techniques, refining self-attention mechanisms, and developing new architectural variants [75], [76]. Recently, researchers have combined ViTs with CNNs, known as HVTs, to merge their benefits [77]. These advancements have facilitated significant improvements in medical image segmentation methods [78], [79].

IV. STRUCTURAL ENHANCEMENTS IN TRANSFORMER-BASED SEGMENTATION

In medical imaging, the segmentation of various cancer cells is essential for helping pathologists make timely diagnosis. ViTs and more recently HVTs due to their compelling benefits over CNNs have emerged as highly effective solutions [80]. ViT and HVT-based systems have shown remarkable performance across various medical image analysis tasks [81] including breast cancer detection [82], [83], [84], COVID-19 detection [85], [86], [87], [88], histopathology image analysis [68], [89], [90], [91], [92], [93], skin lesion detection [94], [95], [96], [97]. These approaches have considerably enhanced the accuracy and efficiency of medical image segmentation and have the potential to enhance clinical diagnosis and decision-making [98], [99].

The following sub-sections categorize these ViT and HVT-based approaches based on their structural modifications.

A. VIT-BASED APPROACHES

Several segmentation-based ViTs have been proposed which can be broadly classified based on their designs. Most of these approaches utilize a UNet-like encoder-decoder architecture, where the ViT architecture is employed in either of the following ways: 1) in the encoder, 2) in the decoder, 3) between the encoder and decoder, or 4) in both the encoder and decoder. These categories are depicted in Figure 2, and their details are discussed in the following sections.

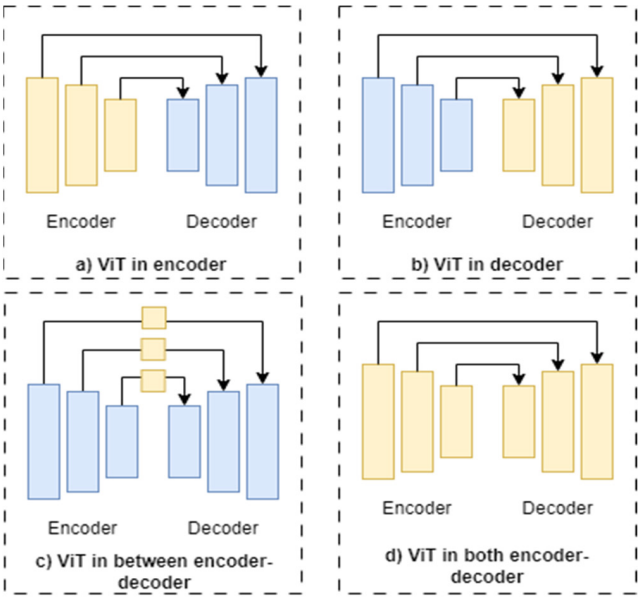


FIGURE 2. Taxonomy of ViT-based medical image segmentation approaches, where yellow-colored layers are the ViT-based layers, blue represent pure CNN layers.

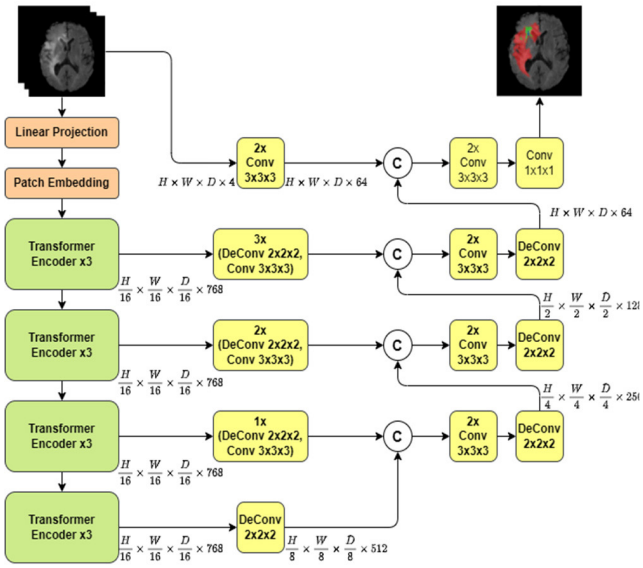


FIGURE 3. Architectural diagram of UNETR, where ViT is utilized in the Encoder architecture.

1) ViT IN ENCODER

An encoder-decoder architecture forms the basis of the most medical image segmentation techniques. The encoder architecture is responsible for learning the hidden embedding

in the images, which are then decoded to a segmentation mask by a decoder. For an effective information transfer between the encoder and decoder, skip connections are also added along with a conventional sequential flow. These skip paths are either direct connections or CNN-based processing blocks.

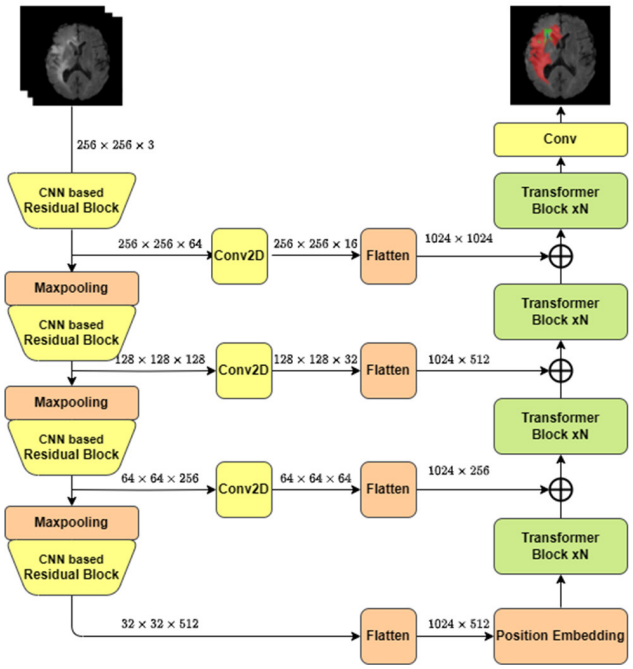


FIGURE 4. Architectural diagram of ConvTransSeg, where ViT is employed in Decoder.

Many ViT-based medical image segmentation approaches utilize a ViT architecture in their encoder to incorporate global relationships in their learned latent space. Such techniques benefit from ViT's MSA to exploit global features during feature extraction in their encoder block. The CNN-based decoder then utilizes this high-level information for predicting pixel-level segmentation masks.

In a study by Hatamizadeh et al, a Transformer-based architecture UNETR (UNet Transformer) is developed to perform 3D medical image segmentation by overcoming the locality-related limitations of UNet [100]. The architectural diagram of UNETR is depicted in Figure 3. It has a ViT-based encoder to effectively capture multi-scale global information from input volumes.

To compute the final semantic segmentation output, the encoder is linked directly to a decoder at different resolutions, via skip connections, similar to a U-Net. They have reported that their model outperformed other CNN-based and Transformer-based models both in terms of computational efficiency and segmentation accuracy.

2) ViT IN DECODER

Some encoder–decoder techniques incorporate a ViT module only in their decoder to generate precise segmentation maps [101].

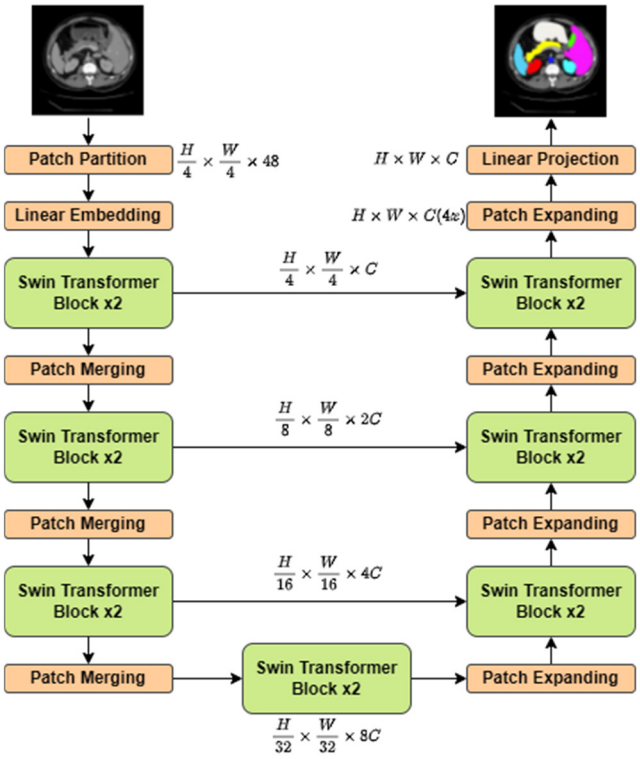


FIGURE 5. Swin-UNet architecture, with VIT in both Encoder-Decoder architectures.

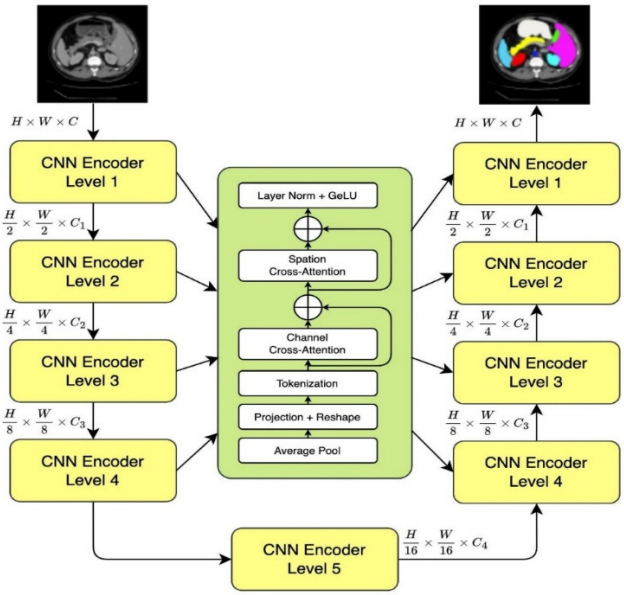


FIGURE 6. Architecture of the DCA Net, with ViT in between Encoder-Decoder.

As the global context is very important in predicting boundary-perfect segmentation masks and differentiating between background and object of interest, the MHSA is beneficial for such an objective during the decoding stage. ConvTransSeg [101] utilizes a CNN-based encoder for feature learning and a ViT-based decoder, interlinked

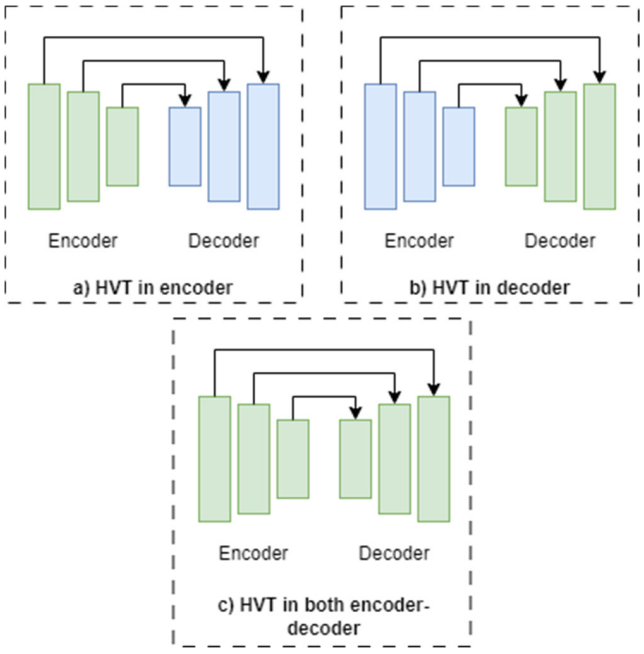


FIGURE 7. Taxonomy of various medical image segmentation approaches based on HVTs, where green-color represents hybrid (CNN + ViT) layers.

at multi-stages to produce the final mask. Figure 4 shows its architectural details. ConvTransSeg showed outstanding results on both binary and multi-class segmentation.

3) ViT IN BOTH ENCODER-DECODER

Various researchers have utilized ViTs in both the encoder and decoder architectures to fully exploit the attention mechanism. Cao et al, introduced Swin-Unet, to perform segmentation in medical images [102]. Its architecture is based on Swin Transformer [103], with the encoder having a shifted-window-based approach and patch embedding layers in the decoder, as shown in Figure 5. Both encoder and decoder architectures are designed hierarchically to enhance segmentation accuracy and robust generalization.

Zhou et al. proposed nnFormer [104], a volumetric Transformer network to carry out segmentation. The architecture utilizes attention-based skip connections to incorporate both local and long-term relationships in the volumetric images. Moreover, they utilized volume-based (V-MSA) and shifted version (SV-MSA) of multi-head self-attention to reduce the computational complexity and capture multi-scale information. MISSFormer [105] proposed by Huang et al., is an encoder-decoder architecture that leverages the specialized blocks for the fusion of hierarchical features with less computational complexity. TransDeepLab [106] combines the DeepLabv3 network [107] with a shifted window-based Swin Transformer. It utilizes variable window sizes within the Swin-Transformer modules to fuse information at multiple scales.

4) ViT IN BETWEEN ENCODER-DECODER

Li et al. proposed ATTransUNet, an architecture based on ViTs utilizing adaptive tokens. Within their model, ViT is integrated into skip connections [108]. They utilized an Adaptive Token Extraction Module (ATEM) in their encoder to extract the most discriminating visual tokens in the images. This results in reduced complexity and enhanced performance. In the decoder architecture, they employed a Selective Feature Reinforcement Module (SFRM) to focus on the most contributing features.

Dual Cross-Attention (DCA), proposed by Ates et al., employs a simple yet effective attention module to enhance the medical image segmentation [109]. Its architecture is depicted in Figure 6. By modeling both channel and spatial relationships from the features derived by the multi-scale encoder, DCA bridges the semantic disparity between the encoder and decoder features.

DCA first employs the Channel Cross-Attention (CCA) and then the Spatial Cross-Attention (SCA) module to model channel-wise and spatial relationships. At last, the encoder features are upsampled to correspond with the corresponding layers of the decoder ViT-V-Net applies ViT blocks at the bottleneck layer between the CNN-based encoder and decoder for volumetric medical image registration task in an unsupervised manner [110]. CoTr provides an efficient hybrid architecture that uses CNN for feature extraction and a deformable self-attention mechanism for global context modeling [111]. The selective multi-scale deformable multi-head self-attention (MS-DMSA) is used, reducing computational complexity and enabling faster convergence.

5) DISCUSSION

Several architectural enhancements in ViT-based models have significantly enhanced their performance in medical image segmentation. By utilizing the MSA mechanism in their architectures, these models effectively capture long-range dependencies that are essential for accurately identifying complex anatomical structures. Additionally, the patch or positional embeddings in ViT-based models enhance feature representation and contextual understanding. This improvement allows the model to better distinguish between relevant and irrelevant patches, ultimately boosting its overall generalization. Furthermore, these enhancements make the model more robust against noise in medical images, which is a significant challenge in this field. However, the limited ability of ViT architectures to capture local relationships within images may affect their efficacy, particularly in medical imaging, where precision and accuracy are essential.

B. HVT-BASED APPROACHES

Pure ViT architectures, which rely solely on attention mechanisms without incorporating convolutional operators, may overlook certain low-level details, potentially impacting segmentation accuracy. In contrast, HVTs utilize the strengths

of both ViTs and CNN architectures, allowing them to capture long-range and local context in the input images. This effective combination enables HVTs to achieve impressive performance across various tasks, particularly in medical image segmentation [112], [113], [114]. These HVT-based segmentation models can be categorized into a) HVT in the encoder branch, b) in the decoder branch, and c) in both the encoder and decoder branches, as shown in Figure 7.

1) HVT IN ENCODER

The most common trend in recent research is to incorporate HVTs in the encoder stage. TransUNet utilizes an HVT architecture in its encoding stage and a cascaded upsampler in its decoding stage [115]. It combines the advantages of ViT and U-Net, capturing both local correlations and long-term relationships within the input images for accurate segmentation results, as shown in Figure 8. Wang et al. proposed a segmentation architecture named TransBTS that combines a ViT and a 3D CNN for global contextual information and local details, respectively [66]. It processes 3D medical volumetric data to capture local correlation and long-range dependencies within the image slices. TransFuse utilized a BiFusion module to combine its encoder and decoder branches which were CNN and ViT architectures, respectively [116].

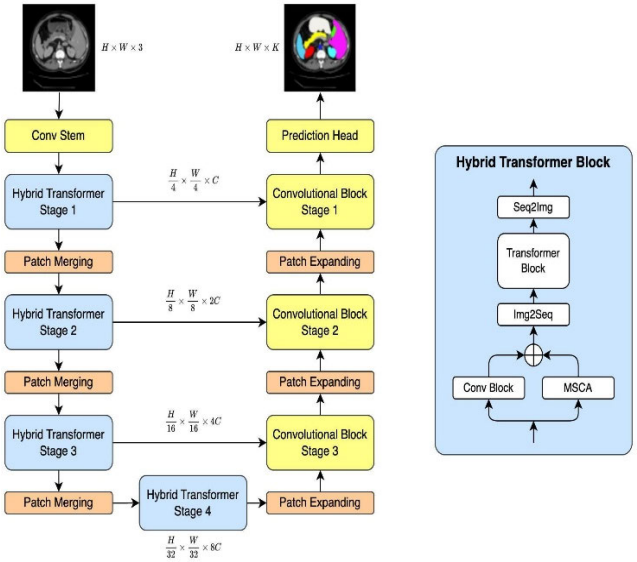


FIGURE 8. Architectural diagram of TransUNet, where a hybrid ViT is utilized in the encoder.

MedT introduced a novel attention mechanism and training strategy for medical image segmentation [117]. It consists of a global module (ViT-based) and a local module (CNN-based) to capture high-level and fine details within the pixels. Swin-UNETR combines Swin Transformers with a U-shape architecture for the task of segmenting brain tumor regions [118]. This entails dividing the input into non-overlapping patches and utilizing windowing mechanisms. However, the integration of the self-attention mechanism into CNNs may result in high computational complexity due to the large spatial size. In this regard, H2Former is introduced to efficiently combine the strengths of MSA and CNNs to perform medical image segmentation [119]. H2Former outperformed the previous techniques by maintaining its computational efficiency in terms of model parameters, Floating Point Operations (FLOPs), and inference time.

2) HVT IN DECODER

To leverage the benefits of HVT in decoding stages, recent research incorporates an HVT in the decoder architecture. A recent study presents a unified framework, UNetFormer, employing a 3D Swin Transformer on the encoding side and a combination of CNN and Transformer on the decoding side [120]. Its architecture is shown in Figure 9.

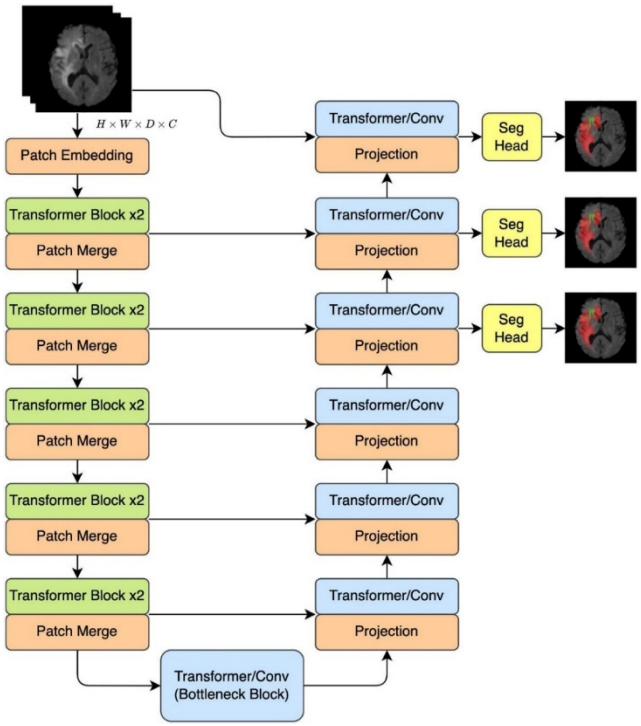


FIGURE 9. Architecture of UNetFormer, where in the decoder architecture where an HViT is employed.

3) HVT IN BOTH ENCODER-DECODER

Many researchers utilize ViT and CNN-based modules in their architecture's encoder and decoder parts [121]. Recently, MaxViT-UNet has been developed as an innovative HVT-based decoder specifically for medical image segmentation [122]. Its architectural diagram is depicted in Figure 10. The authors effectively utilized the multi-axis self-attention mechanism, allowing the model to attend to features along local and global axes, enhancing the discriminative capacity between object and background regions and thus improving segmentation efficiency [123].

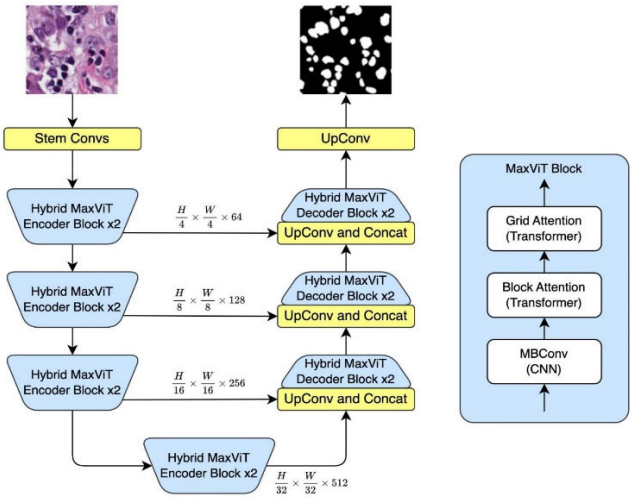


FIGURE 10. Architecture of MaxViT-UNet, with hybrid ViT in both encoder and decoder.

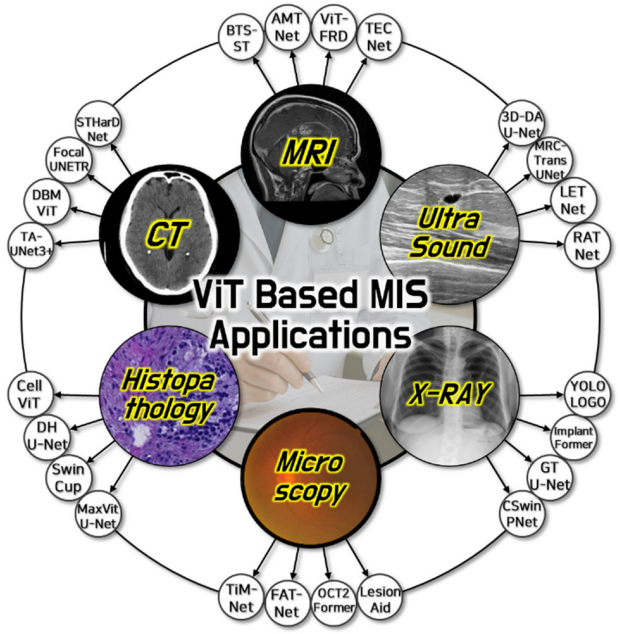


FIGURE 11. Taxonomy for Section V highlighting recent ViT-based models for different modalities.

4) DISCUSSION

With the rise of HVT models in medical image segmentation, researchers have proposed various architectural innovations that incorporate CNN layers into the ViT framework. Some architectures use attention layers only in the encoder, while others apply them in the decoder or across both stages. This flexibility enhances feature representation and helps models adapt to different details in medical images, making them more effective for complex tasks like tumor and organ segmentation. However, this increased complexity can raise interpretability concerns, which are important for deploying these medical systems on edge devices.

V. MODALITY-SPECIFIC ViT-BASED SEGMENTATION APPLICATIONS

Medical image segmentation is a crucial task in medical diagnosis [48], [124], [125], and ViTs have significantly gained attention due to their ability to capture global pixel relationships. HVTs, which are based on both CNNs and ViTs, have also demonstrated notable results in segmenting medical images across diverse modalities [126], [127], [128]. These modalities include but are not limited to CT images, MRI images, ultrasound, X-ray images, histopathological images, and microscopy images (Figure 11). The following subsections review several state-of-the-art ViT and HVT methods across different modalities, with brief discussions and empirical comparisons presented in the corresponding tables.

A. CT IMAGES

Computed Tomography (CT) is an advanced medical imaging technique to produce detailed cross-sectional body images using X-rays for disease detection and diagnosis. Most earlier approaches to CT image segmentation were based on CNNs. However, ViTs have changed this trend [128], [129] and recently many approaches have combined both CNNs and ViTs for improved segmentation in CT images [103], [130], [131], [132], [133].

TAU-Net3+ replaced UNet's CNN encoder with a ViT for kidney tumor segmentation, and a novel attention mechanism called an encoder-decoder Transformer (EDformer) was added to the skip connection to learn local characteristics [129].

In response to the challenges encountered by conventional methodologies in achieving accurate prostate segmentation in CT images, the FocalUNETR introduced an image segmentation framework leveraging a focal Transformer [130]. The developed approach proficiently captured both localized visual features and comprehensive contextual information by incorporating an auxiliary module known as ''boundary-induced label regression''. Their model showed notable performance while segmenting prostate CT images. Hao et al. proposed DBM-ViT, for the discrimination of COVID-19, and other pneumonia types by utilizing CT and Chest X-ray images [131]. The utilization of depth-wise convolutions with diverse expansion rates within DBM-ViT enhanced the ability to capture global information. Recently, Liu et al., developed a cross-shaped windows-based hybrid model to strategically merge local and global features for medical image segmentation in multi-organ CT images [132]. An empirical comparison of these methods is presented in the following section, and in Table 2.

1) EMPIRICAL ANALYSIS OF CT-BASED ARCHITECTURES

The comparison of CT segmentation models reveals that HVTs, particularly those with ViTs in the encoder, demonstrate a favorable trade-off between accuracy and efficiency. For example, TransUNet achieves a high DSC of 0.884 on

TABLE 2. Different ViT and HVT-based techniques for CT image modality.

Method	Architecture	Organ	Dataset	Highlights	Performance Metrics
nnFormer (2023) [105]	HVT (Both in encoder and Decoder)	Multi-organ	Synapse	Uses hybrid architecture to fully leverage the hierarchical and global learning of CNNs and ViTs, respectively.	0.865 DSC
TA-UNet3+ (2023) [130]	ViT (in Encoder and in between Encoder-Decoder)	Kidney, Tumor	KiTS19 Dataset [134]	Integrated Transformer and U-Net in their architecture and proposed an attention mechanism, EDformer, into the skip connection.	0.963 DSC
TransUNet (2024) [135]	HVT	Kidney Liver, Pancreas, Spleen, Gall bladder, Aorta	BTCV [136]	Both CNN and ViT encoders and decoders were used to learn enriched local and global representations.	0.884 DSC
			MSD [137]		0.676 DSC
			Pancreas (private) Dataset		0.697 DSC
FocalUNETR (2023) [131] (code)	ViT (Encoder-based)	Prostate	AMOS Dataset	It adeptly captures both local and global features and incorporates an auxiliary label regression task triggered by boundaries to handle indistinct edges in CT images.	0.8379 DSC
			Private Dataset		0.8923 DSC
Y-CA-Net (2024) [138]	HVT (Hybrid encoder)	Multi-organ	Synapse	It employed a CNN-based local and a ViT-based global branch in its hybrid encoder for enriched feature learning. In addition, they fused the features from both branches using their feature mixer module.	0.870 DSC
			MSD		0.934 DSC

BTCV with only 41.4M parameters and fast inference (0.058s). In contrast, FocalUNETR, which uses a ViT-only encoder, has more parameters (48.3M) and slower inference (5.35s) with a slightly lower performance. Although TA-UNet3+ achieves the highest accuracy (0.963), it focuses on a single organ and does not capture the complexity of multi-organ segmentation Overall, these findings suggest that incorporating HVTs with Transformer modules in the encoder reduces model parameters while maintaining competitive segmentation performance.

B. MRI IMAGES

Magnetic Resonance Imaging (MRI) generates intricate visual representations of tissues and organs within the human body through the application of robust magnetic fields. This imaging modality is particularly valuable in the examination of anatomical regions such as the joints, muscles, and vital organs like the heart and liver, spinal cord, and brain [138].

In recent times, there has been a notable increase in the utilization of ViT-based methodologies for the segmentation of cardiac MRI data, as evidenced by approaches such as [139], [140], [141], [142], [143], and [144]. Fan et al. introduced a knowledge refinement technique called Visual Transformer with Feature Recombination and Feature Distillation (ViT-FRD) [139]. In their approach, ViT functions as the student network, assimilating knowledge from a CNN and acting as the teacher network through optimized distillation losses. ViT-FRD incorporates two enhancements to improve training efficiency and efficacy.

The advancement of an automated and accurate methodology for brain tumor segmentation has the potential to expedite diagnosis across a range of oncologic phenotypes. In recent times, there has been a notable increase in the introduction of CNN layers in the Transformer-based models for the efficient segmentation of brain tumors, as evidenced by several studies [64], [65], [66], [67], [118], [120], [145], [146], [147]. One such model, DenseTrans [148], combines the Swin

Transformer with an enhanced UNet++ network. This integration aims to extract local features from the convolutional layer and global features from the shift window operation and self-attention mechanisms of the Swin Transformer within the high-resolution layer. STHarDNet was developed by Gu et al. to perform cerebral hemorrhage segmentation [149]. In their approach, HarDNet and Swin Transformer were integrated to enhance the segmentation performance beyond what each model can achieve independently. In the case of 3DCAT-BraTs [145], researchers adapted the Swin Transformer for the segmentation of brain tumors in 3D MRI images. This Transformer incorporates a modified CNN-encoder architecture, characterized by residual blocks and a channel attention module. Furthermore, in a different investigation, a robust 3D fusion segmentation network named AMTNet [150] was introduced, building upon the conventional U-shaped structure. Notably, AMTNet incorporates a Transformer-based feature fusion module designed to enhance the integration of multimodality features.

ViTs have been widely utilized in the area of breast lesion segmentation utilizing MRI images, as evidenced by several recent studies [151], [152], [153]. Iqbal et al. introduced BTS-ST [152], inspired by the Swin Transformer, to enhance the feature representation capability, particularly for irregularly shaped tumors. BTS-ST innovatively incorporated pixel-level correlations within the Swin Transformer block to encode spatial knowledge. To address the problem of information loss, a Feature Compression Block was introduced, and a Relationship Aggregation block enabled the cohesive hierarchical combination of global dependencies from the Swin Transformer with features from CNN. In a separate study, Muller Franzes et al. introduced TraBS, enhancing the original SwinUNETR model for breast segmentation in MRI images [153]. TraBS like UNET, strategically utilized different sized kernels and strides in the encoding stages to maintain a consistent depth. They also utilized 1×1 kernels in the deeper layers as bottlenecks. Table 3 summarizes these MRI-based ViT and HVT segmentation models.

1) EMPIRICAL ANALYSIS OF MRI-BASED ARCHITECTURES

An empirical comparison of ViT- and HVT-based methods for MRI segmentation reveals that HVT architectures—incorporating ViT modules in the encoder, decoder, or fusion stages—tend to offer comparable performance in terms of both segmentation accuracy and model complexity. For example, TransConver, an encoder-based HVT model, achieves notable DSCs (0.8173–0.8668) on the BraTS2019 dataset with only 9M parameters. In comparison, Swin UNETR, a ViT-based model, achieves a DSC of 0.8530 on the BraTS2021 dataset but requires 61.98M parameters. UnetFormer performs even better on the same dataset, achieving a high DSC of 0.8880 with a slightly lower parameter count of 58.98M. Similarly, AMTNet, another HVT model, delivers strong segmentation performance (up to 0.9240 DSC on the WT region) with just 10.9M parameters and an inference time of 3 seconds. In contrast, ViT-only models generally require significantly more parameters to achieve similar or marginally better accuracy. These findings suggest that incorporating ViT modules within the encoder of HVT architectures can reduce computational cost while maintaining high segmentation performance, making encoder-based HVTs more practical for MRI applications.

C. ULTRASOUND IMAGES

Ultrasound, or sonography, utilizes high-frequency sound waves to create real-time visualizations of organs and tissues. Extensively used in obstetrics to monitor pregnancies and evaluate fetal development, ultrasound holds a pivotal position in healthcare. Timely identification of breast cancer has shown the potential to reduce fatality rates highlighting the substantial importance of automated breast tumor detection for healthcare practitioners.

Zhu et al. [156] introduced the region-aware Transformer network (RAT-Net), which skillfully integrates information from breast tumor regions at various scales, leading to accurate segmentation. Likewise, Liu et al. [157] devised a hybrid architecture that incorporates Transformer layers into the decoder segment of a 3D UNet, facilitating efficient tumor segmentation in volumetric breast data. The LET-Net architecture [158], introduced recently, combines Transformer and convolutional approaches. Its feature-aligned local enhancement module facilitates the extraction of distinctive local features while maintaining alignment with adjacent-level features. Additionally, a progressive local-induced decoder focuses on recovering high-resolution spatial details through a series of local reconstruction and refinement stages, guided by adaptive reconstruction kernels and enhanced through a split-attention mechanism.

Similarly, MRC-TransUNet [159] introduced an innovative approach by merging Transformer and UNet components. It incorporates a lightweight MR-ViT to bridge the semantic gap and utilizes a reciprocal attention module (RPA) to counteract potential detail loss. A comparison of these common ViT and HVT-based approaches is listed in Table 4.

1) EMPIRICAL ANALYSIS OF ULTRASOUND-BASED ARCHITECTURES

The comparative analysis of ViT and HVT architectures for ultrasound image segmentation shows that HVT models with Transformer components integrated in the encoder or intermediate layers outperform ViT models in both accuracy and structural adaptability. For instance, Cswin-Pnet, which uses a CNN-Swin Transformer hybrid encoder, achieves a DSC of 0.8725, while 3D-UNet with decoder-level Transformers achieves a comparatively lower score of 0.7636. Similarly, BTS-ST, an encoder-based HVT, yields a better F1-score of 0.9080 by effectively learning spatial irregularities in tumor boundaries. Though exact parameter values and inference times are not reported for most models, RAT-Net, with 64.07M parameters, does not significantly outperform

TABLE 3. Different ViT and HVT-based techniques for MRI image modality.

Method	Architecture	Organ	Dataset	Highlights	Performance
BiTr-UNet (2021) [65] (code)	HVT (in between Encoder-Decoder)	Brain	BraTS2021	Used two Transformer layers and an attention component to improve features from the encoding and decoding units.	DSC: 0.8874 (ET) DSC: 0.9257 (WT) DSC: 0.9350 (TC)
VT-UNet (2022) [66] (code)	ViT (in both Encoder-Decoder)	Brain	BraTS2021	Utilized two ViT layers in the encoder along with the window-based and cross-attention modules to learn global and local relationships.	DSC: 0.8220 (ET) DSC: 0.9190 (WT) DSC: 0.8720 (TC)
Swin-UNETR (2021) [119] (code)	ViT (Encoder-Based)	Brain	BraTS2021	Utilized Swin Transformer in the encoding unit and a CNN in the decoder.	DSC: 0.8530 (ET) DSC: 0.9270 (WT) DSC: 0.8760 (TC)
TransBTS (2021) [67] (code)	HVT (in Encoder)	Brain	BraTS2019	Used both CNN and ViT architectures in their encoder to capture local and global contextual learning.	DSC: 0.7893 (ET) DSC: 0.9000 (WT) DSC: 0.8194 (TC)
			BraTS2020		DSC: 0.7873 (ET) DSC: 0.9009 (WT) DSC: 0.8173 (TC)
SwinBTS (2022) [155] (code)	HVT (in both Encoder-Decoder)	Brain	BraTS2019	SwinBTS utilized a 3D Swin Transformer for local and global-level feature learning along with the convolutional operations for up and down sampling.	DSC: 0.7443 (ET) DSC: 0.8975 (WT) DSC: 0.7928 (TC)
			BraTS2020		DSC: 0.7736 (ET) DSC: 0.8906(WT) DSC: 0.8030 (TC)
			BraTS2021		DSC: 0.8321 (ET) DSC: 0.9183 (WT) DSC: 0.8475 (TC)
UnetFormer (2022) [121] (code)	HVT (Decoder-Based)	Brain	BraTS2021	Utilized 3D Swin Transformer-based encoder and hybrid decoders. They linked both the encoder and decoder via skip connections at five different resolutions.	DSC: 0.8880 (ET) DSC: 0.9322 (WT) DSC: 0.9210 (TC)
CATBraTs (2023) [146]	HVT (Encoder-Based)	Brain	BraTS2021	Proposed 3D Transformer based on Swin Transformer with a modified CNN-encoder architecture.	DSC: 0.8260 (ET) DSC: 0.8760 (WT) DSC: 0.7990 (TC)
AMTNet (2023) [151]	HVT (in both Encoder-Decoder)	Brain	BraTS2021	Introduced Transformer-based feature fusion module designed to better fuse multimodality features.	DSC: 0.7340 (ET) DSC: 0.9240 (WT) DSC: 0.8950 (TC)
DenseTrans (2023) [149]	ViT (in between Encoder-Decoder)	Brain	BraTS2021	Combined Swin Transformer with the improved UNet++ network.	DSC: 0.8830 (ET) DSC: 0.9320 (WT) DSC: 0.8620 (TC)
TransConver (2022) [147]	HVT (Encoder Based, in between Encoder-Decoder)	Brain	BraTS2018	Proposed a parallel module, which uses convolution blocks and Transformer blocks, respectively, to extract local and global information.	DSC: 0.8173 (ET) DSC: 0.9157 (WT) DSC: 0.8668 (TC)
			BraTS2019		DSC: 0.7840 (ET) DSC: 0.9019 (WT) DSC: 0.8257 (TC)

TABLE 3. *(Continued.)* **Different ViT and HVT-based techniques for MRI image modality.**

Focal Cross-Transformer (2023) [148]	ViT (Encoder Based)	Brain	BraTS2021	They utilized cross windows to increase the receptive field for capturing global-level coarse details. In addition, incorporated a focal attention mechanism to refocus on the local level fine details.	DSC: 0.8728 (ET) DSC: 0.9328 (WT) DSC: 0.8735 (TC)
Segtran (2021) [156] (code)	ViT (in between Encoder-Decoder)	Brain	BraTS2019	Incorporated squeeze and excitation attention blocks in their Transformer architecture to focus on the important regions.	DSC: 0.7400 (ET) DSC: 0.8950 (WT) DSC: 0.8170 (TC)
BTS-ST (2023) [153]	HVT (Encoder Based)	Breast	MRI Dataset	Incorporated Swin-Transformer with CNN-based U-Net to enhance the spatial learning of the network.	F1-Score: 0.8330
TraBS (2023) [154]	ViT (Encoder Based)	Breast	UKA	Modified SwinUNETR and added deep supervision for the lower-resolution layers.	DSC: 0.9160
			DUKE		DSC: 0.8640
STHarDNet (2023) [150]	ViT (in between Encoder and Decoder)	Brain	ATLAS Dataset (189 MRI Scans)	Combined the HarDNet and Swin Transformer to propose a hybrid model for cerebral hemorrhage segmentation using 2D CT images.	DSC: 0.7280 IoU: 0.6120

other HVT models despite its complexity, indicating potential redundancy in using heavier ViT modules across multiple layers. These results support the insight that encoder-based HVTs provide an optimal balance between performance and efficiency, making them more suitable for ultrasound applications, where real-time inference and model compactness are often critical.

D. X-RAY IMAGES

X-ray images are important both for diagnosing and treating a patient as it provide important information regarding the interior organs of the human body. Modern deep learning models including ViTs and HVTs can be effectively used to diagnose a variety of medical diseases using X-ray images [162], [163], [164].

Tooth root segmentation is an important stage in dental image analysis as it allows dentists to precisely measure the size and shape of tooth roots and discover any abnormalities that may be present. Many recent studies have employed ViTs for tooth root segmentation [165], [166], [167]. Yang et al. introduced ImplantFormer, an implant position regression network built on Transformers that uses oral CBCT data to predict implant location automatically [166]. ImplantFormer uses a 2D axial image of the tooth crown area to analyze the implant position. In another study [168], Sheng et al. introduced Swin-UNet, a U-shaped Transformer-based architecture with an encoder, decoder, and skip-connections, designed specifically for panoramic radiograph segmentation.

Mammography is a specific type of X-ray used to examine breast tissue for signs of breast cancer. Certain research studies have directed their efforts toward refining architectural designs for improved segmentation of irregular tumor boundaries [160], [169]. CSwin-PNet was recently proposed for the task of breast lesion segmentation. Its architecture is based on a Pyramid Network that integrates the CNN and Swin Transformer [160]. A summary of these methods is provided in Table 5.

1) EMPIRICAL ANALYSIS OF X-RAY-BASED ARCHITECTURES

In X-ray image segmentation, especially for dental and breast applications, recent ViT and HVT models show that integrating Transformer components in both encoder and decoder stages can enhance accuracy, but often at the cost of increased model complexity. For instance, GT U-Net, an HVT model using both CNN and ViT components, achieves a high DSC of 0.9254 for tooth root segmentation, despite using no pre-trained weights. In contrast, ViT-only models such as ImplantFormer and YOLO-LOGO report relatively lower detection performance (e.g., AP@0.75 = 0.1370 and mAP = 0.6500) while relying on large parameter counts like YOLO-LOGO's 77M. Meanwhile, SWin-UNet, another ViT-based model, shows varied performance across datasets (F1-score: 0.6372 on PLAGH-BH vs. 0.8203 on PDX), suggesting sensitivity to data variability. These results imply that HVTs with Transformer modules in both encoder and decoder can potentially generalize better across tasks, while pure ViT models may require careful architectural tuning and

TABLE 4. Different ViT and HVT-based techniques for Ultrasound image modality.

Method	Architecture	Organ	Dataset	Highlights	Performance
RAT-Net (2022) [157] (code)	Hybrid ViT (Encoder Based, in between Encoder-Decoder)	Breast	ABUS	Captured multi-scale information for breast cancer segmentation by utilizing a region-aware Transformer network.	Accuracy: 0.8134% DSC: 0.64 IoU: 0.549
3D-UNet (2021) [158]	Hybrid ViT (in between Encoder-Decoder)	Breast	ABVS	Adopted a structure based on 3D U-Net with attention mechanism and Transformer layers to optimize the extracted image features.	DSC: 0.7636 IoU: 0.6214
Cswin-Pnet (2023) [161]	Hybrid ViT (Encoder Based, in between Encoder-Decoder)	Breast	Dataset1 [162]	Connected CNN and Swin Transformer as a feature extraction backbone to construct a pyramid-structured network for effective feature encoding and decoding.	DSC: 0.8725
			Dataset2 [162]		DSC: 0.8368
BTS-ST (2023) [153]	Hybrid ViT (Encoder Based)	Breast	BUS	Integrated the Swin Transformer into a conventional CNN-based U-Net to improve the feature learning ability for irregularly shaped tumors.	F1-Score: 0.9080

dataset alignment. Moreover, hybrid models often outperform ViT-only models with fewer parameters, supporting the conclusion that HVTs are more parameter-efficient and robust choices for X-ray image segmentation tasks.

E. HISTOPATHOLOGICAL IMAGES

Histopathological images, obtained from biopsy tissue samples, are analyzed by pathologists to investigate cellular architecture and spot microscopic abnormalities or illnesses [171]. Histopathology-based analysis is crucial for timely cancer diagnosis and prevention. While the use of ViTs and HVTs on histopathology images is still in its early stages, nevertheless, they have already given promising results on various histopathology segmentation tasks [172], [173], [174].

Wang et al. introduced DHUnet, an innovative feature fusion approach that blends global and local features for the segmentation of WSI. This method incorporates Swin Transformer and ConvNeXt modules within a dual-branch hierarchical U-shaped architecture [175].

In another recent approach, the Swin Transformer with Cascaded UPsampling (SwinCup) was introduced to segment histopathology images [174]. To improve the feature aggregation of the proposed SwinCup, the authors introduced a cascaded up-sampling decoder in conjunction with the encoder.

In another technique, MaxViT-UNet, the authors proposed an Unet-like encoder-decoder CNN-Transformer framework [122]. The proposed hybrid decoder utilized Multi-Axis Self-Attention (Max-SA) to improve the segmentation results.

These models subsequently have the potential to contribute to patient stratification in clinical trials, patient selection for targeted therapy, and personalized treatment planning. Table 6 enlists the main contributions of these histopathology-based ViT and HVT segmentation approaches.

1) EMPIRICAL ANALYSIS OF HISTOPATHOLOGICAL IMAGES-BASED ARCHITECTURES

ViT architectures have also demonstrated strong performance in histopathological image segmentation, particularly through hybrid vision transformer (HVT) architectures that integrate both CNN and ViT components. Models such as DHUnet and MaxViT-UNet effectively combine local and global feature representations, achieving high DSC and Jaccard scores across multiple datasets while maintaining a manageable number of parameters. For instance, MaxViT-UNet achieves a DSC of 0.8378 with only 24.72 million parameters, whereas CellViT—a ViT-only encoder model with 699 million parameters—achieves a slightly lower F1-score of 0.8300 on the PanNuke dataset. These results suggest that strategically incorporating Transformer modules within CNN-based architectures can yield comparable segmentation performance with significantly lower computational cost. Moreover, HVTs, especially those leveraging ViTs in both the encoder and decoder, offer a more efficient and generalizable framework for histopathology image analysis.

F. MICROSCOPY IMAGES

Microscopy involves using microscopes to view small objects, such as cells, tissues, and microorganisms [177]. Different types of microscopy, such as light microscopy, electron microscopy, and fluorescence microscopy, offer various levels of detail and resolution [178], [179]. The skin lesions segmentation is a critical step in computer-aided diagnosis

TABLE 5. Different ViT and HVT-based techniques for X-Ray image modality.

Method	Architecture	Organ	Dataset	Highlights	Performance
GT U-Net (2021) [166] (code)	Hybrid ViT (in both Encoder-Decoder)	ToothRoot	Tooth Root Segmentation Dataset	Utilized both the CNN and Transformer architectures without pre-training weights.	DSC: 0.9254
			DRIVE		ACC: 0.9631
ImplantFormer (2022) [167]	ViT (Encoder-Based)	ToothCrown	Private dataset	ViT-based regression model is developed to predict the implant position in CT tooth images.	AP@0.75: 0.1370
SWin-UNet (2023) [169]	ViT (in both Encoder-Decoder)	Tooth	PLAGH-BH (Private dataset)	Implemented SWin-Unet CNN for tooth segmentation on panoramic radiographs.	F1-score: 0.6372
			PDX [171]		F1-score: 0.8203
YOLO-LOGO (2022) [170]	ViT (Encoder-Based)	Breast	CBIS-DDSM	A two-branched architecture to detect and isolate breast masses in mammograms.	mAP: 0.6500
			INBreast		mAP: 0.6140

TABLE 6. Different ViT and HVT-based techniques for histopathological image modality.

Method	Architecture	Organ	Dataset	Highlights	Performance
CellViT (2023) [68] (code)	ViT (Encoder-Based)	Multi-tissue dataset	PanNuke	Devised a ViT-based model to segment cell nuclei in digitized tissue samples.	F1-score: 0.8300
			MoNuSeg		F1-Score: 0.851
			CoNSeP		AUROC: 0.963
DHUnet (2023) [176] (code)	HVT (Encoder-Based)	Liver, Lung, Breast	Liver Dataset	Employed a Unet-based architecture with a Swin Transformer-based branch and a ConvNeXt branch to integrate both local and global features to segment Whole Slide Images (WSI).	DSC: 93.07 Jaccard: 87.04
			WSSS4LUAD		DSC: 86.54 Jaccard: 77.87
			BCSS		DSC: 78.88 Jaccard: 68.01
MaxViT-UNet (2023) [123] (code)	HVT (in both Encoder and Decoder)	Multi organs	MoNuSeg18	The MaxViT-UNet leverages the hybrid MaxViT block in all encoder and decoder stages. The MaxViT block, consisting of MBConv, Block Attention and Grid Attention, effectively extracts local and global features.	DSC: 0.8378 IoU: 0.7208
			MoNuSAC20		DSC: 0.8215 IoU: 0.7030

and therapy planning. It enables the objective identification and measurement of lesion size, shape, and characteristics, playing a vital role in classifying benign and malignant lesions, early detection, and monitoring temporal changes. Recent methods have leveraged ViTs for skin lesion segmentation, inspired by their success in medical imaging. [94], [95], [96]. Notably, in the LesionAid framework [96], the authors introduced a novel multi-class prediction approach for skin lesions, employing ViT and ViTGAN [180]. To mitigate issues related to class imbalance, ViT-based Generative Adversarial Networks (GANs) are employed in these techniques.

Alterations in the structure and function of retinal blood vessels have been associated with cardiovascular conditions, including coronary heart disease, atherosclerosis, and hypertension. The automated segmentation of retinal images and subsequent analysis are essential for the evaluation and prediction of related diseases, thereby contributing to public health. Various research approaches have effectively harnessed the capabilities of Transformers in the context of retinal vessel segmentation in prior studies [181], [182], [183], [184], [185], [186], [187].

The OCT2Former [184], characterized by an encoder-decoder architecture, employs a dynamic Transformer encoder in conjunction with a lightweight decoder. The dynamic token aggregation Transformer within the dynamic Transformer encoder is instrumental in capturing global context information about retinal vessels, while an auxiliary convolution branch addresses the Transformer's inherent lack of inductive bias. Du et al. introduced an ensemble strategy that integrates different foundational deep learning models, such as the FCN-Transformer and Pyramid Vision Transformer (PVT), for retinal vessel segmentation [182]. Main contributions of these microscopy-based segmentation models are presented in Table 7.

1) EMPIRICAL ANALYSIS OF MICROSCOPY-BASED ARCHITECTURES

In microscopy image segmentation both HVT and ViT architectures have shown comparable results. However, in some cases HVT models have outperformed ViT models even with a less numbers of parameters, particularly in skin lesions and retinal vessels. MTPA-UNet, an encoder-based HVT, achieves high accuracy (97.18% on DRIVE) with only 11.8 million parameters, making it much more efficient than models like PCAT-UNET (57.4 million) and OCT2Former (49.99 million). Similarly, Axial Transformer + CNN and GT U-Net exploit hybrid strategies to achieve high F1 and accuracy scores without relying on pretraining or large architectures. Critical comparison of microscopy-based models reveals that the utilization of both HVTs and ViTs in encoders can result in better performance both in terms of segmentation accuracy and computational complexity.

G. DISCUSSION

Latest advancements in ViT and HVT models have shown promising results in medical image segmentation across various modalities, such as CT, MRI, ultrasound, X-ray, histopathology, and microscopy [188]. ViT-based models have outperformed traditional CNNs in various tasks like brain tumor segmentation using MRI scans [65], achieving 0.871 DSC as compared to 0.664 DSC for CNNs. This achievement is credited to their capacity to model long-range dependencies, which is crucial for segmenting large-scale structures like tumors.

However, ViTs have faced challenges in modalities like ultrasound, where their global attention mechanism struggles to model fine-grained spatial details due to the image's lower resolution and noise. HVT models, which combine ViTs with CNNs, have addressed this limitation by capturing both local and global information. For example, CSwinPNet, an HVT model, has achieved DSC of 0.87, surpassing both pure ViT (0.805) and CNN (0.778) models while segmenting breast lesions in ultrasound images.

Despite these improvements, the complexity of medical images across different modalities continues to pose significant challenges. Further research is needed to refine these models to achieve more robust segmentation results, ultimately aiming to provide better assistance to pathologists in clinical decision-making.

VI. RECENT TRENDS AND CHALLENGES

ViTs and HVTs have shown great potential for revolutionizing medical imaging, offering an effective approach to analyzing and extracting valuable data from medical images. Their potential to learn intricate spatial relations and global context make them suitable to solve complex medical imaging problems. However, there are several open challenges that need to be addressed to fully harness their capabilities in medical image segmentation.

One major limitation of ViTs is that, unlike CNNs, they rely heavily on self-attention mechanisms, which tend to overlook fine-grained local details. HVTs aim to mitigate this by capturing both global and local information simultaneously. However, scalability and high computational demands pose significant constraints especially when dealing with large and detailed 3D medical images, requiring model modifications, parallelization, and optimized decoding techniques [21].

Another major challenge is the limited availability of medical image datasets, often due to imbalanced classes and differences in imaging methods. Current trends to mitigate these include data augmentation, synthetic data generation, and advanced training techniques like self-supervised learning to make the models more robust [189], [190], [191], [192], [193]. However, generalizability and robustness of these methods remain challenging, especially in unseen cases or rare conditions.

Precise annotations are critical for training effective medical image segmentation models, but they tend to be time intensive and susceptible to human bias. Few-shot and zero-shot learning techniques may offer promising alternatives by reducing the reliance on extensive labeled datasets [194], [195]. Although recently a number of studies have also employed active learning to overcome the dependency on large labeled datasets [196], however, further research is needed to improve the reliability and adaptability of these methods across diverse clinical applications.

Recently, federated learning is utilized in various medical image segmentation approaches to deal with the privacy concern of medical data [197], [198]. In federated learning, the model is trained on the decentralized data without sharing sensitive information. Despite its promising results, it also

TABLE 7. Different ViT and HVT-based techniques for microscopy image modality.

Method	Architecture	Organ	Dataset	Highlights	Performance
FAT-net (2022) [95] (code)	Hybrid ViT (Encoder-Based)	Skin lesion	ISIC 2016	Implemented a dual encoder comprising both CNNs and Transformer branches to simultaneously gather local characteristics and global context information.	DSC: 0.9159
			ISIC 2017		DSC: 0.8500
			ISIC 2018		DSC: 0.8903
			PH2		DSC: 0.9440
Fully Transformer Network (2022) [96] (code)	ViT (in both Encoder-Decoder)	Skin lesion	ISIC 2018	Offered a self-attention paradigm for image segmentation and other downstream tasks.	DSC: 0.9000 Jaccard: 0.829
CAViT (2022) [127]	ViT (in between Encoder-Decoder)	Eye (Retinal vessels)	DRIVE	Introduced channel attention vision Transformer that jointly applies the efficient channel attention and the vision Transformer.	Accuracy: 0.9700%
			CHASE DB1		Accuracy: 0.9761%
TiM-Net (2022) [188] (code)	ViT (in between Encoder-Decoder)	Retinal images	STARE	Integrated M-Net, various attention mechanisms, and weighted side output layers to create TiM-Net for the segmentation of retinal vessels.	Accuracy: 0.9711% AUC: 0.9670
			CHASE DB1		Accuracy: 0.9711% AUC: 0.9648
			DRIVE		Accuracy: 0.9638% AUC: 0.9682
PCAT-UNET (2022) [182]	Hybrid ViT (in both Encoder-Decoder)	Retinal images	DRIVE Dataset	Proposed an encoder-decoder network leveraging both ViT and Convolution and incorporating skip connections to merge deep and shallow features from both modules.	F1-score: 0.8160 Accuracy: 0.9622%
			CHASE DB1		F1-score: 0.8273 Accuracy: 0.9812%
			STARE Dataset		F1-score: 0.8836 Accuracy: 0.9796%
OCT2Former (2023) [185]	Hybrid ViT	Retinal images	OCTA-SS	Introduced OCT2Former, employing an encoder-decoder structure that	Jaccard: 0.8344

TABLE 7. *(Continued.)* **Different ViT and HVT-based techniques for microscopy image modality.**

	(Encoder-based)		ROSE-1	comprises a dynamic Transformer encoder and a lightweight decoder primarily.	Jaccard: 0.7855
			OCTA-6M		Jaccard: 0.8099
			OCTA-3M		Jaccard: 0.8513
Axial Transformer and Convolutional Neural Network (2022) [125]	Hybrid ViT (Encoder-based)	Retinal images	CHASE DB1	Both the Axial Transformer and CNN architectures were used to simultaneously extract the global and local relationships in the images.	F1-score: 0.8095 Accuracy: 0.9657%
MTPA_Unet (2022) [184]	Hybrid ViT (Encoder-based)	Eye (Retinal images)	DRIVE	The Transformer is fused with the CNN, and the TPA module is used to learn both local and global-level information.	Accuracy: 0.9718% DSC: 0.8318
			CHASE DB1		Accuracy: 0.9762% DSC: 0.8164
			STARE		Accuracy: 0.9773% DSC: 0.8557
Group Transformer Network (GT U-Net) (2021) [166] (code)	Hybrid ViT (in both Encoder-Decoder)	Eye (Retinal images)	DRIVE	Utilized both the convolution and Transformer networks without pre-training weights.	F1-score: 0.8458

faces challenges like data differences, communication costs, and maintaining consistent performance. Diffusion models are also being explored for generating realistic synthetic data to improve segmentation [199], [200]. However, integrating them with ViTs and HVTs is still being researched. Several innovative pre- and post-processing techniques are also developed to refine segmentation results. Additionally, understanding how ViTs make decisions is important for clinical use, but it's still difficult due to interpreting the diffused attention maps. Addressing these key challenges and developing new techniques to enhance ViT and HVT-based medical image segmentation can further accelerate progress in the healthcare community.

VII. FUTURE RECOMMENDATIONS

Dealing with the specific challenges faced by ViTs in medical image segmentation presents several avenues for future research and innovation.

One key direction involves the development of new hybrid architectures by combining the strengths of ViTs and CNNs. This hybrid approach aims to strike a balance between capturing global structural relationships and local details crucial for precise medical imaging segmentations, facilitating the deployment of real-world automated diagnostic systems.

Improving training paradigms and methodology for ViTs in medical applications is another focus area. Fine-tuning on public datasets, incorporating relevant constraints, and employing adaptive methods to address biases and variations in class distributions are crucial steps. Additionally, complex reinforcement learning algorithms and tailored data augmentation techniques can enhance ViTs' capabilities to handle diverse pathological properties within medical images.

Exploring ensemble methods emerges as a fascinating approach, involving the combination of predictions from multiple diverse ViT backbones. By leveraging various models with different specifications and architectures, this collaborative approach aims to enhance resilience, versatility, and overall efficiency in medical visualization tasks, reducing reliance on individual networks.

Incorporating domain knowledge is emphasized to make deep learning more effective, especially in medical imaging. Techniques such as attention mechanisms can be fine-tuned to focus on critical areas, and the integration of anatomical priors or handcrafted features can guide the model. This incorporation of domain knowledge from medical professionals enhances segmentation accuracy and reliability in medical image analysis.

Transfer learning stands out as a promising approach to address data scarcity challenges. Pre-training ViTs on large-scale natural image datasets and fine-tuning them on specific medical imaging tasks allows the models to leverage knowledge from diverse images. This approach enables ViTs to generalize better, improve performance, and potentially reduce the need for extensive labelled medical datasets.

Lastly, multi-modal learning is identified as an exciting opportunity for ViTs in medical image segmentation. ViTs, with their ability to handle sequence data, can effectively accommodate inputs from various modalities, capturing dependencies and interactions between different data sources. This multi-modal capability enhances the accuracy and reliability of medical image segmentation by incorporating complementary information, ultimately aiding clinicians in making more effective diagnosis.

VIII. CONCLUSION

ViT-based image segmentation techniques have shown predominant performance in a number of image-related applications, including medical images. The self-attention mechanism in ViTs enables the model to learn global relationships in the images. This paper presents a detailed discussion of several architectural modifications along with some latest trends and training techniques utilized to improve the performance of ViT-based medical image segmentation approaches.

Despite ViTs' limitation in capturing local correlations in medical images, researchers have introduced approaches combining ViTs and CNNs to capture both local and global perspectives efficiently. We also discuss HVT-based medical image segmentation techniques in detail and categorize them based on their placement in encoder-decoder-based medical image segmentation architectures. In addition, we present a modality-specific overview of how these ViTs and HVTs are used in various real-world medical image segmentation applications, showcasing their impact on diagnosis, treatment, and disease monitoring.

Medical image analysis using ViTs is rapidly advancing, but there are still challenges to address, such as training effective models with limited medical data, understanding how these models make decisions, and customizing them for specific medical tasks. By overcoming these challenges and combining ViTs with other advanced techniques, we can significantly improve medical image analysis and ultimately provide better care for patients.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors used ChatGPT to improve the grammar in the text. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

DATA AVAILABILITY

No data were used for the research described in the article.

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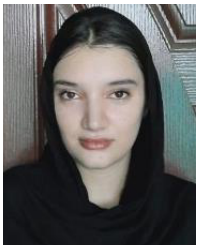


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